Answers: Introduction to factorials, permutations, and combinations

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Summary

Answers to questions relating to the guide on introduction to factorials, permutations, and combinations.

These are the answers to Questions: Introduction to factorials, permutations, and combinations.

Please attempt the questions before reading these answers!

Q1 Factorials

1.1.

(a) There are 120 ways of listing the 5 different types of bonbons.

(b) There are 5040 ways of listing the 7 different types of jelly sweets.

(c) There are 24 ways of listing the 4 different types of fudge.

1.2. There are 20922789888000 ways of listing all 16 different types of sweets.

1.3. There are 87091200 ways of listing the 16 different types of sweets, if you want to keep all sweets of a certain type together.

Q2 Permutations

2.1.

(a) If there are 6 types of gumdrops sold at Lovelace's Lollies, but only 3 spots on the shelf for these, there are 120 ways you could organize the sweets.

(b) If there are 7 types of liquorice, but only 2 spots on the shelf for these, there are 42 ways you could organize the sweets.

(c) If there are 10 types of fizzy sweets, but only 6 spots on the shelf for these, there are 151200 ways you could organize the sweets.

2.2. Combining all 3 of these types of sweets, you can organize the 21 different kinds of sweets into the 11 available spots in 41508446720000 different ways.

2.3. If you want to include at least 2 of each type, there are 980755776000 ways to organize the sweets.

Q3 Combinations

3.1.

(a) If there are 8 types of lollies sold at Cantor's Confectionery, there are 28 ways you could pick 2.

(b) If there are 4 types of chocolate, there are 4 ways you could pick 3.

(c) If there are 12 types of marshmallow, there are 494 ways you could pick 4.

3.2. Combining all 3 of these different types of sweets, there are 346104 ways you could pick 7.

3.3. Assuming you don't want all 7 to be marshmallows or lollies, there are 345304 ways to pick.

Q4 Counting Strategies

4.1. If Lovelace's Lollies sells 3 red sweets, 3 yellow sweets, and 2 blue sweets, and you want to arrange them without having all 3 red sweets together, there are 36000 ways to do this. That is $8! - (3! \cdot 6!)$, all possible permutations, take away the permutations where the red sweets are together.

4.2. For a display in Cantor's Confectionery, if you want to arrange the 8 bonbons in a circle, there are 5040 ways to do this.

4.3. If you want to stock 3 kinds of lollies and 4 kinds of jelly sweets, making sure no 2 lollies are next to each other, there are 1440 ways to do this.

4.4. If there are 10 different types of sweets, including 5 kinds of fizzy sweets, and you want to arrange these sweets on the shelf, but you want to keep all the fizzy sweets together, there are 86400 ways to do this.

4.5. If there are 12 different types of sweets, 4 of which are fudge, and you want to pick 6 sweets, including at least 2 different types of fudge, there are 1260 ways to do this.

Version history and licensing

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